

Statement of Teaching Suitability

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Overview

Between 2015 and 2023, I taught mathematics at university level in Brazil, primarily at the Federal University of Viçosa (UFV) in Minas Gerais, with an earlier course conducted at the University of Brasília. Across this period I taught calculus, linear algebra, abstract algebra, and number theory to undergraduate students across roughly twenty course offerings spanning eight academic years, in addition to a two-week intensive analysis course during a summer school. This statement summarises that teaching record and sets out how I approach the classroom, in order to support applications where teaching ability, course design, and student mentoring are part of the required profile.

Teaching Record

My teaching spans four main areas of the undergraduate mathematics curriculum:

- **Calculus:** Calculus I and Calculus II, taught in nearly every semester from 2016 through 2023 — the course I have taught most consistently and to the largest number of students.
- **Linear Algebra:** Notions/Introduction to Linear Algebra, taught in multiple semesters between 2016 and 2019.
- **Abstract Algebra:** Rings and Modules (2016), Algebra I and Algebra II (2017–2020), giving me repeated experience teaching structural, proof-based mathematics to students encountering it for the first time.
- **Number Theory and Discrete Mathematics:** a dedicated course on Additive Number Theory at the University of Brasília (2015), and later Finite Mathematics and Fundamentals of Arithmetic at UFV (2022–2023) — courses that draw directly on my own doctoral specialisation in combinatorial number theory.

I also taught several special-topics and advanced-problem courses (Tópicos Especiais em Matemática II and III, Problemas Especiais II, 2018–2019) and a two-week intensive Analysis course as part of a summer school (2017) — formats that required condensing material into a shorter, high-intensity teaching arrangement rather than a standard semester-long structure.

Teaching Philosophy

My approach to teaching mathematics is shaped directly by my own research background: a doctorate in combinatorial number theory taught me that rigour and intuition are not in tension, and that the clearest explanations usually come from finding the right concrete example before presenting the general argument. In the classroom, I consistently try to give students a foothold — a specific case, a picture, a small computation — before asking them to reason abstractly, particularly in proof-heavy subjects like algebra and number theory where students often struggle with the transition from computation to structure.

Teaching the same core courses (especially Calculus I) across many consecutive semesters gave me the opportunity to iterate: to notice recurring points of confusion and adjust how I introduced them the following term, and to build problem sets that anticipated common errors rather than only testing for them after the fact. Teaching a wide range of levels — from first-year calculus to graduate-adjacent special-topics courses — has also made me comfortable calibrating explanations to very different levels of mathematical maturity within the same institution, a skill that has carried over directly into my later work explaining technical machine learning and statistical methods to non-specialist collaborators.

Relevance Beyond the Classroom

Since moving from academia into applied data science and machine learning research in Germany, I have continued to rely on the communication and mentoring skills built during this teaching period — whether presenting technical results to interdisciplinary research partners, writing documentation intended for non-specialist users, or explaining statistical and machine learning methods to collaborators from other domains. I see teaching, supervision, and this kind of applied technical communication as closely related skills, developed over the same eight years and continuing to inform how I work today.

Closing

I would welcome the opportunity to bring this teaching background to a role that values course design, student supervision, or technical mentoring alongside research, and I am glad to provide further detail, a full list of course offerings, or references from colleagues and students at UFV on request.